

Reinforcement Learning Project Report

Snow plow problem with multi-agent reinforcement learning

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1 Problem Description: Snow Plow Problem

We consider a snow plow problem where a fleet of snow plows is tasked with clearing snow from a network of roads in a city. The objective is to minimize the total time taken to clear all roads while ensuring that the snow plows operate efficiently and avoid collisions. Each snow plow can be controlled by an agent that makes decisions based on the current state of the environment, including the location of other snow plows, the amount of snow on each road, and the traffic conditions.

2 Environment

A grid represents the city, where each cell corresponds to a road segment. The state of the environment includes the positions of all snow plows, the amount of snow on each road segment, and any obstacles or traffic conditions that may affect the movement of the snow plows.

As an initial environment size we consider a grid of size 15×15 with a maximum of 2 snow plows. The snow plows can move in four directions (up, down, left, right) and can also choose to stay in their current position. The environment is dynamic, with snow accumulating on the roads over time, and the agents must adapt their strategies accordingly.

2.1 Observations

2.1.1 Experiment 1: 1-Agent + Full Observation

2.1.2 Experiment 2: 1-Agent + Partial Observation

PO-range: (3×3) or (5×5)

2.1.3 Experiment 3: 2-Agents + Full Observation

2.1.4 Experiment 4: 2-Agents + Partial Observation

PO-range: (3×3) or (5×5)

2.2 Actions

1. $\rightarrow$: Move Right
2. $\leftarrow$: Move Left
3. $\uparrow$: Move Up
4. $\downarrow$: Move Down
5. Do nothing: Stay in the current position

2.3 Truncation Dynamics

Episode ends if the city streets are cleared or if a maximum number of time steps is reached. The truncation dynamics ensure that the agents learn to optimize their actions within a limited time frame, promoting efficient snow clearing strategies.

- After 1000 time steps, the episode ends.
- If all roads are cleared before 1000 time steps, the episode ends early.

3 Theoretical Part

References